Teach, Check, Respond

Session 1 — Estimate with Benchmarks · Session 2 — Estimate a Percent of a Number · Maryland 6.AT.4

LESSON ROUTE

about 45 minutes per session

Warm up · 5 min	Warm-Up page (Session 1). Note who needs the prerequisite sheet.
Launch · 7 min	Slides: name the target. Students predict an answer to the Model, Reason, Explain task.
Model · 10 min	Guided Notes page for the session. Draw the deck's model next to the worked example.
Practice · 18 min	Practice Form A: core #1, #2, #4, #6 and Find the Error. Add more from what you see.
Check · 5 min	Exit Ticket. Sort into got it / almost / need help with the table below.

Shorter period: keep the model and the exit ticket; trim practice. Form B is for reteach or retake, so keep it unseen until then.

SESSION 1 · ESTIMATE WITH BENCHMARKS

I can estimate a percent by comparing an amount to benchmark percents.

EXIT TICKET – WHAT TO LOOK FOR AND WHAT TO SAY NEXT

#	Expected answer	Next prompt if it is not there yet
1	a. 50% b. 75%	Between which two benchmarks is 31 out of 40? Which one is it closer to? Use 10% of 40 to check.
2	25% of 300 = 75, and 74 is close to 75, so about 25%.	About what percent of the 300 beads are NOT used? Show the benchmark amount.
3	**C** (50%)	Is 19 out of 40 a little more or a little less than 50%? How do you know?

THE ERROR TO WATCH FOR

Writes the part as the percent without using the whole (82 out of 120 called "82%"), or picks a benchmark without finding the benchmark amounts first.

Reteach Draw a bar for the whole, have the student label the 25%, 50%, and 75% amounts, and ask them to point to where the part falls before naming a percent.

- Using the wrong whole (calling 30 GB of music 75% because 150 of 200 GB is used).
- Writing the part as the percent (47 out of 64 called "47%").

Calculation slip only: use Session 1 Prerequisite Practice A, then go back to the model. Wrong whole or wrong model: label the whole (100%) before redoing the arithmetic.

FRESH CHECK – GIVE AFTER RETEACHING (STUDENTS HAVE NOT SEEN IT)

Check	A garden has 56 tomato plants. 13 of them have ripe tomatoes. About what percent have ripe tomatoes? Name the benchmark amount you used.
Answer	25% of 56 = 56 ÷ 4 = 14. 13 is close to 14, so about 25% (exact: about 23%).

SCORE THE CHECK

3	Model fits the numbers, correct answer with units, explanation matches. Move on; remove a support next time.
2	Right setup with a small slip, or a thin explanation. Quick feedback, then one more new problem.
1	Some correct facts, but the wrong whole or model. Reteach with the session's model and the prerequisite page.
0	Not enough to tell yet. Ask the student to point, draw or say it before writing.

LANGUAGE ACCESS

Point and draw, then say it with a partner ("Partner A: "The whole is ___. The part is close to ___, which is ___% of the whole." Partner B: "How do you know it is closer to ___% than to ___%?""), then write one sentence (Explain Your Thinking sheet), then explain alone. Fade one support each time.

SESSION 2 · ESTIMATE A PERCENT OF A NUMBER

I can estimate a percent of a number using friendly percents and rounded amounts.

EXIT TICKET – WHAT TO LOOK FOR AND WHAT TO SAY NEXT

#	Expected answer	Next prompt if it is not there yet
1	a. about 10 (20% of 50) b. about 30 (75% of 40)	Is your estimate for 76% of 41 high or low? Which rounding tells you?
2	40% of $60: 10% = $6, so 40% = about $24.	Now estimate the sale price of the backpack. Is it a little high or a little low?
3	**C** (80)	Estimate 20% of 390 a second way, using 10% first. Which estimate is closer?

THE ERROR TO WATCH FOR

Believes there is just one right estimate and erases a reasonable one (says 140 is wrong for 26% of 588 because a classmate got 150), or moves the decimal the wrong way when finding 10%.

Reteach Put two estimates side by side, ask which rounded numbers each one used, and decide together which is closer to the exact answer and why.

- Misplacing the decimal when finding 10% (10% of 300 written as 3).
- Multiplying by 20 instead of 0.20 for 20% (20 × 500 = 10,000).

Calculation slip only: use Session 2 Prerequisite Practice A, then go back to the model. Wrong whole or wrong model: label the whole (100%) before redoing the arithmetic.

FRESH CHECK – GIVE AFTER RETEACHING (STUDENTS HAVE NOT SEEN IT)

Check	Estimate 49% of 818 in two ways. Name the compatible (friendly) numbers you used each time.
Answer	Examples: 50% of 800 = 400; 50% of 820 = 410. Exact: about 401. Accept any close estimate with named compatible numbers.

SCORE THE CHECK

3	Model fits the numbers, correct answer with units, explanation matches. Move on; remove a support next time.
2	Right setup with a small slip, or a thin explanation. Quick feedback, then one more new problem.
1	Some correct facts, but the wrong whole or model. Reteach with the session's model and the prerequisite page.
0	Not enough to tell yet. Ask the student to point, draw or say it before writing.

LANGUAGE ACCESS

Point and draw, then say it with a partner ("Partner A: "I used ___% of ___ because ___." Partner B: "I used ___% of ___. Our estimates are different because ___.""), then write one sentence (Explain Your Thinking sheet), then explain alone. Fade one support each time.